

Mehmet Aytuğ Yürük

Undergraduate Researcher in Computer Vision & Medical Imaging
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Education

B.Sc. in Artificial Intelligence & Data Engineering GPA: 3.79 / 4.00 <i>Istanbul University</i>	2024 – 2028 (Expected) <i>Istanbul, Turkey</i>
High School Diploma <i>Sakarya Science High School</i>	2020 – 2024 <i>Sakarya, Turkey</i>

Research Experience

Undergraduate Researcher <i>Istanbul University</i>	2025 – Present <i>Istanbul, Turkey</i>
- Fine-tuned and evaluated five ResNet and four Vision Transformer architectures across two retinal age-estimation studies on 9,857 images, using ImageNet-pretrained models and consistent patient-disjoint data splits while investigating Graham filtering as a retinal preprocessing strategy; achieved a best test MAE of 4.86 years with ViT-L/16. - Explored alternative color-space representations for retinal age prediction through single-model experiments, a 12-channel fusion architecture, and matched ensemble comparisons; the color-space ensemble reduced MAE by 3.06% over its RGB counterpart, supported by 100,000 patient-cluster paired bootstrap resamples. - Evaluated BCE, Asymmetric Loss, and hybrid objectives for long-tailed multi-label concept recognition on RO-COV2 (79,789 radiology images, 1,947 UMLS concepts); ASL improved the EfficientNet-B0 score from 0.5809 to 0.5882, with the best configuration reaching 0.5916 using EfficientNet-B4.	

Publications & Manuscripts

- M. A. Yürük**, A. Memiş. “Decoding Chronological Age from the Retinal Fundus Images: A Deep Learning-based Analysis with Vision Transformers.” *34th Signal Processing and Communications Applications Conference (SIU)*, 2026. IEEE. **Published**.
- M. A. Yürük**, A. Memiş. “Age Prediction and Categorization from Retinal Fundus Images Using Residual Neural Networks.” *5th International Informatics and Software Engineering Conference (IISEC)*, 2026. IEEE. **Published**.
- M. A. Yürük**, A. Memiş. “Multi-Label Medical Concept Recognition using Asymmetric Loss-Driven EfficientNet: Analyzing the Detection of Rare Concepts Under Long-Tailed Imbalance.” *MEDICON 2026*. **Accepted**.
- M. A. Yürük**, A. Memiş. “Exploring the Impact of Alternative Color Spaces in Deep Retinal Age Prediction from Fundus Images.” *Submitted to ATEEC 2026*. **Under Review**.
- F. Ç. Eraslan, K. E. Eslemmez, **M. A. Yürük**, et al. “Calibration and Selective-Risk Reliability of Tabular Models under Controlled Feature Shift.” *UBMK 2026*. **Accepted**.

Selected Projects

BiyoVes – Local Portrait-Preparation App

- Developed an open-source local Python app for Turkish biometric (50 × 60 mm) and vesikalık (45 × 60 mm) photo preparation.
- Uses MediaPipe Face Landmarker for eye-line correction and BiRefNet Lite for background matting, then composes 300-DPI white-background outputs. Supports single-photo and folder processing.

Technical Skills

Python, PyTorch, OpenCV, Git, GitHub

Awards & Competitions

TEKNOFEST 2242 University Students Research Project Competition Finalist	2025
TÜBİTAK 2204-D Climate Change Research Projects Competition Two-time Finalist	2022, 2023